

CE 310 — Week 2 Lab: Shape Matters — Reading Distributions

Dataset: Tucson Airport Hourly Weather, 2023 (same file as Week 1) Topic: Histograms
· Skewness · Quartiles · Percentiles

Before You Begin

Files needed: - Week1_Tucson_Weather_2023.csv — same file from Week 1 - Week2_Answer_Sheet.xlsx — download from D2L and enter all answers here

Answer Sheet column map:

Column	What to enter
Col B	(pre-filled question label — do not edit)
Col C	Your Excel formula or method description
Col D	★ Intermediate value (★ questions only — leave blank otherwise)
Col E	Your numerical answer
Col F	(pre-filled unit — do not edit)

Your answer cell matches the question label — A1 goes in row A1 of Col C, B3 in row B3.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761). Without the apostrophe Excel evaluates it and your method is lost. Screenshots are not accepted; the typed formula is what earns the approach credit.

Excel Quick Reference

Goal	Formula / Method
Count values in bin	=COUNTIFS(B2:B8761, ">="&lo, B2:B8761, "<"&hi) — primary method
Bin count, other convention	=FREQUENCY(data, bin_edges) — Ctrl+Shift+Enter (older Excel) or Enter (365)
Build histogram chart	Insert → Statistical Charts → Histogram (Excel 2016+, Mac & Windows)
Build histogram (alt)	Select COUNTIFS table → Insert → Column Chart → Gap Width = 0%

Goal	Formula / Method
ToolPak histogram (opt.)	Data → Data Analysis → Histogram — Windows only
Skewness coefficient	=SKEW(B2:B8761)
First quartile (Q1)	=QUARTILE(B2:B8761, 1) or =PERCENTILE(B2:B8761, 0.25)
Third quartile (Q3)	=QUARTILE(B2:B8761, 3) or =PERCENTILE(B2:B8761, 0.75)
Nth percentile	=PERCENTILE(B2:B8761, n/100) — e.g., 0.90 for 90th pct
Monthly average	=AVERAGEIF(E2:E8761, 8, D2:D8761) — month 8 = August
Count above threshold	=COUNTIF(B2:B8761, ">100")

This week you'll build the tools to read a distribution's shape, not just its center.

Setup (~10 min)

1. Re-open your Week 1 Excel file, or re-import Week1_Tucson_Weather_2023.csv as a new workbook.
2. Confirm your column layout:
 - Column A: DateTime
 - Column B: Temp_F
 - Column C: Wind_Speed_mph
 - Column D: Relative_Humidity_pct
 - Column E: Month (from Week 1 Setup — =MONTH(A2), filled to row 8761)
 - If column E is missing, add it now: in E1 type Month, in E2 type =MONTH(A2), then fill down to E8761.
3. No add-ins required for Section A. This week's histogram method uses built-in Excel formulas that work on both Mac and Windows without any add-in. The Analysis ToolPak is listed as an optional alternative at the end of the histogram section if you prefer it.
4. Quick calculation — Variance: Before starting Section A, compute the temperature variance in any blank cell:

=VAR(B2:B8761)

You should get approximately 320.8 °F². Note the relationship: VAR = STDEV². Your Week 1 A5 computed STDEV ≈ 17.91°F, and 17.91² = 320.8. Variance and

standard deviation describe the same spread — variance has squared units ($^{\circ}\text{F}^2$), which are hard to interpret directly, so engineers almost always report standard deviation instead. Week 2 introduces skewness, which captures something variance and stdev cannot: the asymmetry of the distribution.

Section A — Histogram Basics (~25 min)

Bin counts → whole numbers. SKEW coefficients → 2 decimal places. CV (A9) → 4 decimal places.

Counting Histogram Bins

Method 1 — COUNTIFS formula (all versions, Mac & Windows — use this):

COUNTIFS counts rows that satisfy two conditions simultaneously — a lower bound and an upper bound. For each bin, change the boundary numbers:

Bin	Formula
60–70°F	=COUNTIFS(B2:B8761, ">="&60, B2:B8761, "<"&70)
70–80°F	=COUNTIFS(B2:B8761, ">="&70, B2:B8761, "<"&80)
80–90°F	=COUNTIFS(B2:B8761, ">="&80, B2:B8761, "<"&90)

The pattern is: lower boundary is $\geq$ (inclusive), upper boundary is $<$ (exclusive). Build out all bins from 20°F to 120°F in a column — you can fill them down by changing just the two numbers.

Method 2 — FREQUENCY() formula (all Excel versions, alternative):

=FREQUENCY(B2:B8761, bins_array)

- In a blank column, enter the bin upper edges: 20, 30, 40, ..., 120 (one value per row)
- Excel 2019 and earlier: select a range of cells one row taller than your bin column, type the formula, then press Ctrl+Shift+Enter (array formula)
- Excel 365: just press Enter — the result spills automatically

Method 3 — Analysis ToolPak (Windows only, optional): 1. Data tab → Data Analysis → Histogram → set Input Range to $\$B\$1:\$B\8761 , Bin Range to your bin-edge column, check Labels and Chart Output → OK. 2. Not available on Mac Excel without extra setup — use Method 1 or 2 instead.

Building the Histogram Chart

Method 1 — Insert → Statistical Charts (Excel 2016+, Mac & Windows — recommended):

1. Select your COUNTIFS bin counts (with labels): the two-column table of bin labels and

counts you built above. 2. Insert tab → Charts group → Statistical Charts icon → Histogram. 3. Excel draws the chart from your frequency table automatically. 4. Right-click the bars → Format Data Series → set Gap Width to 0% so the bars touch (standard histogram convention).

Method 2 — Column chart from your COUNTIFS table (all Excel versions): 1. Select your bin-label and count columns → Insert → Column Chart → Clustered Column. 2. Right-click → Format Data Series → Gap Width = 0%. 3. This produces an equivalent chart from your COUNTIFS frequency table.

A1 — Temperature Histogram: 60–70°F Bin

Using the COUNTIFS method above, find the count of hours where temperature was in the 60–70°F range ($\geq 60^\circ\text{F}$ and $< 70^\circ\text{F}$).

A2 — Temperature Histogram: 80–90°F Bin

Find the count of hours in the 80–90°F range.

This is the modal bin — the most common temperature range in Tucson’s 2023 data.

A3 — Verify with FREQUENCY()

Use the FREQUENCY function (Method 2 above) to build the full bin table, then find the row for the 70–80°F bin. With upper edges as your bin column, that is the row labeled 80.

Watch the bin convention — the two methods will not match. COUNTIFS as written in Method 1 counts $\geq$ lower and $<$ upper, the interval $[70, 80)$. FREQUENCY counts the other way: greater than the previous edge and less than or equal to the current edge, the interval $(70, 80]$. Hours sitting exactly on a boundary land in different bins, and on this dataset the two counts differ by more than 100 hours. That gap is the lesson, not a mistake in your work. Report your COUNTIFS count for the 70–80°F bin as your A3 answer, and note in Col C which convention you used.

A4 — Temperature Skewness

Calculate the skewness coefficient for the temperature column using SKEW.

Interpretation note: SKEW returns a number near zero here — the coefficient confirms the near-symmetric shape. Compare: your A2 (modal bin at 80–90°F) vs. the mean ($\sim 72^\circ\text{F}$). The mode is well above the mean, which is the hallmark of left skew — but at only -0.05 , the skewness is barely detectable numerically.

A5 — Wind Speed Histogram: 0–5 mph Bin

Build a second frequency table for wind speed (column C) using the same COUNTIFS approach. Bins: 0–5, 5–10, 10–15, 15–20, 20–25, 25–30, 30–35 mph.

Find the count of hours in the 0–5 mph bin.

A6 — Wind Speed Histogram: 5–10 mph Bin

From your wind speed frequency table, find the count of hours in the 5–10 mph bin.

Notice: Is the 5–10 mph bin larger than the 0–5 mph bin? What does this tell you about where the “typical” wind speed falls?

A7 — Wind Speed Skewness

Use SKEW on the wind speed column.

Positive SKEW means a right-skewed distribution — the bulk of values cluster at low speeds with a long tail of occasional high winds. Compare the magnitude here to the temperature SKEW in A4.

A8 — Humidity Skewness

Use SKEW on the relative humidity column.

A9 — Coefficient of Variation (CV)

The coefficient of variation (CV) expresses standard deviation as a fraction of the mean: $CV = \sigma / \mu$. It is dimensionless, so you can compare variability across datasets with different units or scales. Use STDEV and AVERAGE on the temperature column.

A CV of ~0.25 means the standard deviation is about 25% of the mean — substantial relative variability. Compare: San Francisco annual temperature has a CV well below 0.10 (mild, stable climate), while Tucson’s desert swings push CV above 0.24. When $CV > 0.20$, designing for the average misses the tails by a large margin.

Section B — In-Depth Analysis (~40 min)

Report temperatures in °F, speeds in mph, percentages in %.

B1 — Temperature Q1 (Verify with PERCENTILE)

You computed Q1 in Week 1 (answer A10 $\approx 58^\circ\text{F}$). This week, use the PERCENTILE function — introduced in the Week 2 tutorial — to confirm the same value.

PERCENTILE(range, p) takes a decimal argument between 0 and 1. This returns the same result as =QUARTILE(B2:B8761, 1), confirming that Q1 is simply the 25th percentile.

B2 — Temperature Q3 (Verify with PERCENTILE)

Verify your Week 1 Q3 (answer A11 $\approx 86^\circ\text{F}$) using PERCENTILE at the 75th percentile.

B3 ★ — Temperature IQR

IQR = Q3 – Q1

Col D: Q1 (from B1) · Col E: IQR = Q3 – Q1

Why PERCENTILE matters beyond Q1/Q3: =QUARTILE() only works for Q1, Q2, Q3. =PERCENTILE() handles any arbitrary threshold — B4 and B5 below use the 90th and 95th percentiles, which QUARTILE cannot return. That is the reason Week 2 introduces PERCENTILE.

B4 — Temperature 90th Percentile

Use PERCENTILE on the temperature column at the 90th percentile.

B5 ★ — Temperature 95th Percentile

Use PERCENTILE on the temperature column at the 95th percentile.

Col D: 90th percentile (from B4) · Col E: 95th percentile

B6 — Hours Above the 95th Percentile

How many hours in 2023 had a temperature strictly above your B5 value? Use COUNTIF on the temperature column with your B5 value as the threshold.

Connection to Week 1: This count should match your Week 1 A7 answer (hours $> 100^\circ\text{F}$). That's not a coincidence — your B5 value IS the 95th percentile, meaning exactly ~5% of hours (≈ 438) fall above it. The small discrepancy between the theoretical 5% count and your actual count reflects how PERCENTILE interpolates with discrete data.

Engineering Context: ASCE 7-22 requires structural wind design based on a 50-year return period wind speed — a threshold far more extreme than the 95th percentile of a single year.

B7 — Wind Speed Q1

Use QUARTILE or PERCENTILE on the wind speed column.

B8 — Wind Speed Q3

Use QUARTILE or PERCENTILE on the wind speed column.

B9 — Wind Speed IQR

$IQR = Q3 - Q1$

Compare the wind IQR to the temperature IQR (B3). Which variable has more spread in its middle 50%? Does this surprise you given the SKEW values?

B10 — Wind Speed 95th Percentile

Use PERCENTILE on the wind speed column at the 95th percentile.

B11 ★ — Wind Pressure at 95th Percentile

Using the simplified ASCE wind pressure formula from Week 1 ($C_d = 1.3$):

$$P = 0.00256 \times 1.3 \times V^2$$

where V is your B10 value.

Col D: V^2 (B10 squared) · Col E: P in psf

Compare this pressure to the pressure at the annual average wind speed (~7.24 mph):

$$P_{\text{avg}} = 0.00256 \times 1.3 \times 7.24^2 \approx 0.174 \text{ psf}$$

The ratio shows how much pressure a structure experiences at the 95th percentile wind vs. the average wind — and why average-based design fails.

ArcE Framing: ASHRAE 62.1 requires HVAC latent load sizing to outdoor design humidity — the 90th percentile humidity, not the annual average.

B12 — Humidity 90th Percentile

Use PERCENTILE on the relative humidity column at the 90th percentile.

Target shape for Chart 1

Figure 1: Target shape for Chart 1

B13 — August Average Humidity (Peak Monsoon)

August is Tucson's peak monsoon month. Calculate the average relative humidity for August only, using AVERAGEIF with the Month helper column (column E, month = 8).

B14 ★ — Probability Bridge: $P(80^{\circ}\text{F} \leq T < 90^{\circ}\text{F})$

Using your histogram results, estimate the probability that a randomly selected hour in 2023 had a temperature between 80°F and 90°F :

$$P = (\text{count of hours in } 80\text{--}90^{\circ}\text{F bin}) / (\text{total hours}) \times 100$$

Col D: count of $80\text{--}90^{\circ}\text{F}$ hours (your A2) · Col E: probability as % (e.g., 18.24 for 18.24%)

Required Charts

Insert both charts in your working Excel file.

Chart 1 — Temperature Histogram (7 pts)

Build a histogram of Temp_F (B2:B8761) with 10°F bins. Use Insert → Statistical Charts → Histogram (Excel 2016+), or build a clustered column chart from your COUNTIFS table with Gap Width = 0%.

- Title: Hourly Temperature Distribution — Tucson, AZ 2023
- Axes: Temperature ($^{\circ}\text{F}$) on x, Number of Hours on y
- Add a text box marking Mean $\approx 72^{\circ}\text{F}$ near the $70\text{--}80^{\circ}\text{F}$ bar

Does this histogram show one peak or two? A second distinct peak would signal mixed subpopulations requiring separate analysis — a key EDA diagnostic.

Chart 2 — Temperature Box-and-Whisker Chart (8 pts)

Build a box plot of Temp_F (B2:B8761). In Excel 2016+: Insert → Statistical Charts → Box and Whisker. Older Excel: build a clustered column from a 5-row Min/Q1/Median/Q3/Max summary table with data labels.

- Title: Temperature Box Plot — Tucson, AZ 2023
- Y-axis: Temperature ($^{\circ}\text{F}$)
- Add a text box with IQR = [your B3 value] $^{\circ}\text{F}$

Target shape for Chart 2

Figure 2: Target shape for Chart 2

Does the median sit near the center of the box? Do any outlier dots appear outside the whiskers?

Memo Deliverable

Type your memo directly in the Written sheet of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed. Use the template below.

To: CE 310 Instructor

From: [Your Name], [Your Major]

Date: [Date]

Subject: Week 2 Lab — Distribution Shape Analysis, Tucson 2023 Weather Data

Your three paragraphs each map to an EDA goal: Paragraph 1 → Describe (center, spread, shape); Paragraph 2 → Discover (cross-variable patterns); Paragraph 3 → Check (design conditions).

Paragraph 1 — Temperature Distribution Shape: Describe the shape of the temperature distribution using your histogram (Chart 1) and SKEW coefficient (A4). Reference your box plot (Chart 2) — state where the median falls relative to Q1 and Q3, and what the box width (IQR) tells you about the spread of typical annual temperatures. Cite at least one specific number (bin count, quartile value, or SKEW coefficient). Explain why the shape looks like a plateau rather than a bell curve.

Paragraph 2 — Comparing Variables: Compare the temperature distribution (A4 SKEW) to the wind speed distribution (A7 SKEW) and humidity distribution (A8 SKEW). Contrast the physical meaning: why is wind speed right-skewed while temperature is nearly symmetric? Mention at least one skewness coefficient value.

Paragraph 3 — Latent cooling load and humidity design conditions:

Using your B12 (humidity 90th percentile) and B13 (August average humidity), explain the latent cooling load challenge for a Tucson office building. Compare these values to the annual average humidity (32.67%). Describe what happens to indoor air quality and occupant comfort if the HVAC system is sized for the annual average instead of the 90th percentile condition. Reference ASHRAE 62.1 and the concept of design conditions.

Submission: Save as CE310_W2_<NetID>.xlsx (e.g., CE310_W2_abc123.xlsx) and upload to the Week 2 D2L dropbox. A 5-point deduction applies if your file cannot be matched to your student record.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet
 - ☐ File saved as .xlsx and named CE310_W2_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in Col E — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in Col C, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in Col D
 - ☐ Uploaded to D2L before the deadline
-

Grading Summary

Component	Points
Section A — A1–A9	18
Section B — B1–B14	60
Chart 1 — Temperature histogram	7
Chart 2 — Temperature box-and-whisker chart	8
Memo (3 paragraphs)	7
Total	100
